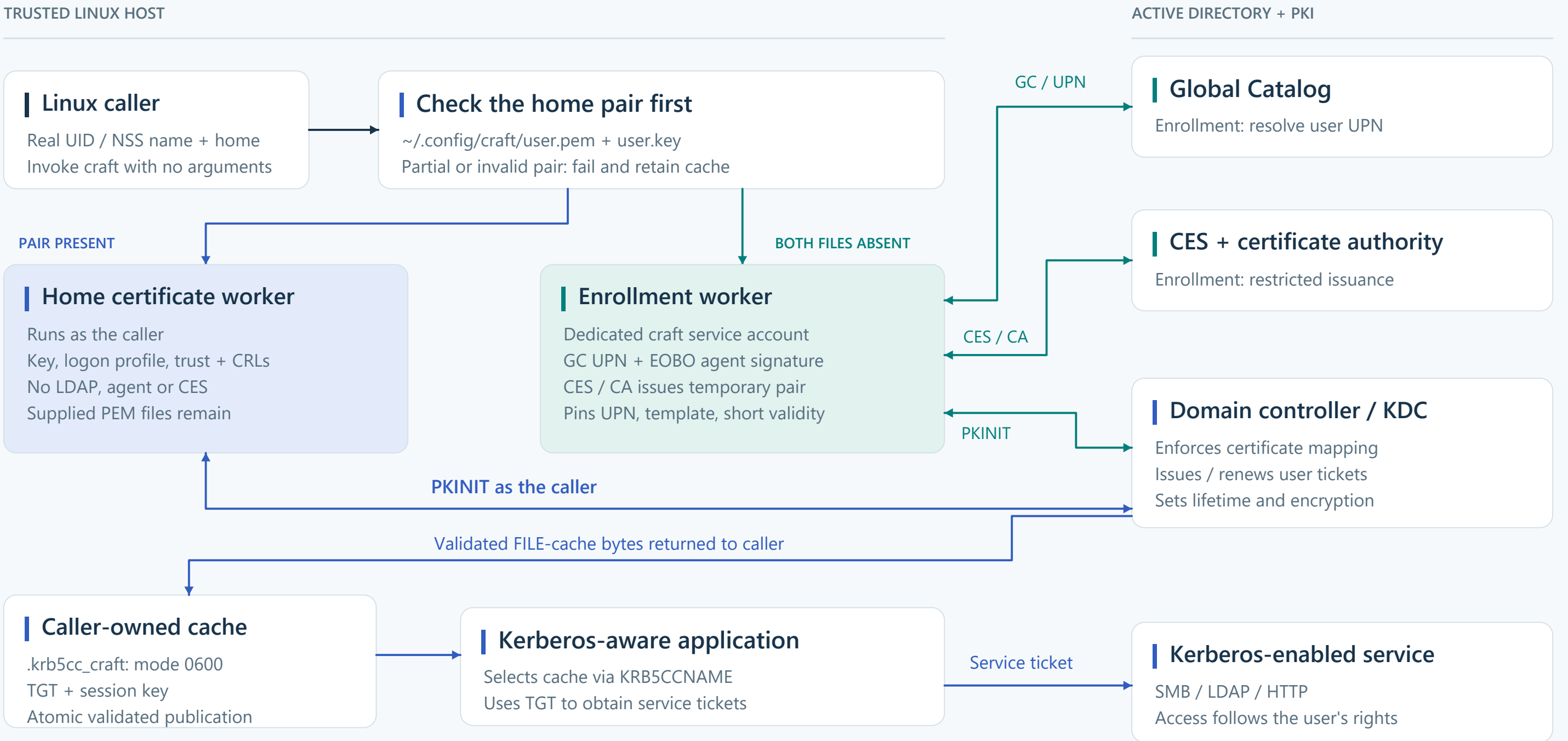


Two certificate paths to one user ticket cache

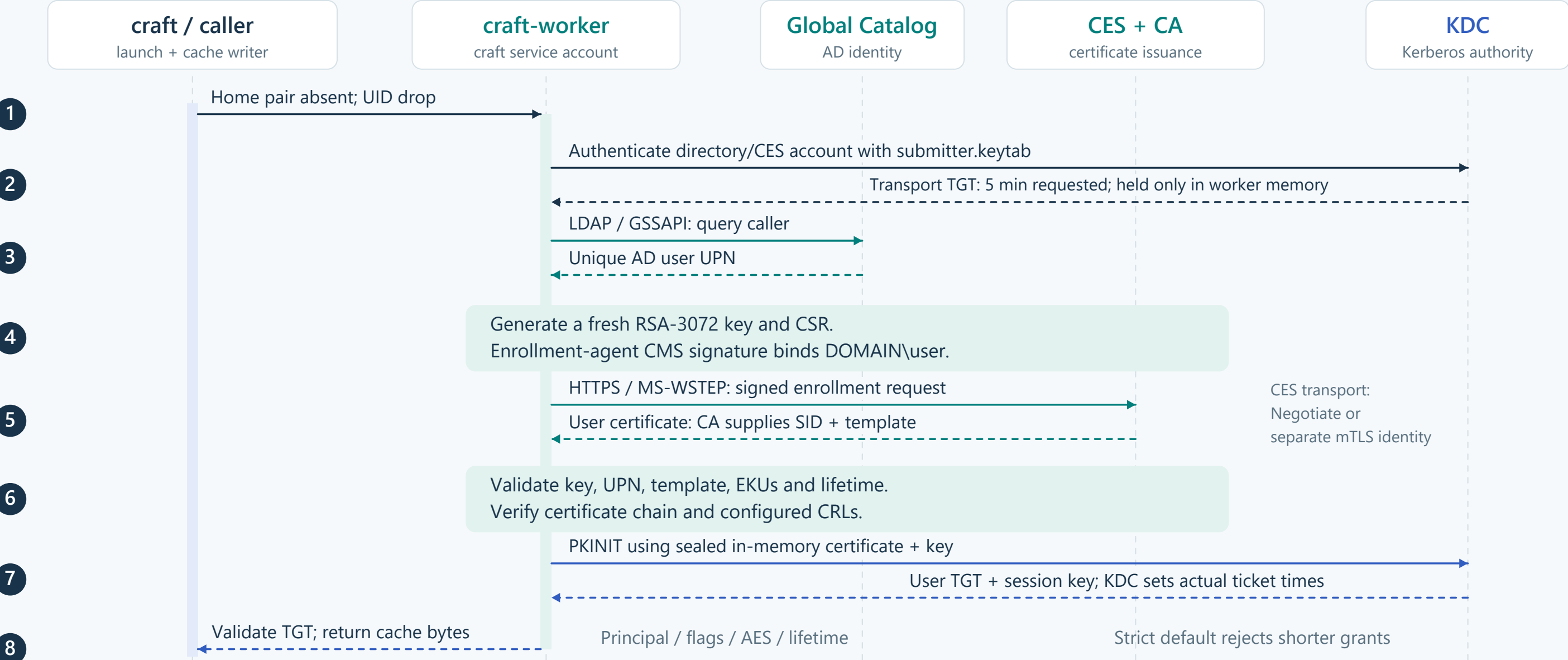
A supplied home pair takes priority. Only an absent pair selects administrator-authorized enrollment.



Public policy, trust and CRLs stay root-controlled. Home-only executables need no setuid or service account.

Enrollment when no home certificate pair is present

Enrollment path only: an absent home pair selects the service account, directory lookup and CES.



Publish only after successful validation

Caller writes a mode-0600 staging file, then atomically replaces .krb5cc_craft. Output is the FILE: cache name.

Ticket clocks and certificate storage

Both paths request 10 h / 7 d by default. Certificate lifetime and KDC policy constrain the initial grant.



Keep a job authenticated, then stop with it.

craft-maintain runs as the caller, coordinates one shared cache per UID, and watches a live process.

START ONCE / WATCH A SHELL THAT STAYS ALIVE FOR THE JOB

```
job_pid=$$
cache=$(/usr/local/bin/craft-maintain --watch-pid "$job_pid") || exit "$?"
export KRB5CCNAME="$cache"
```

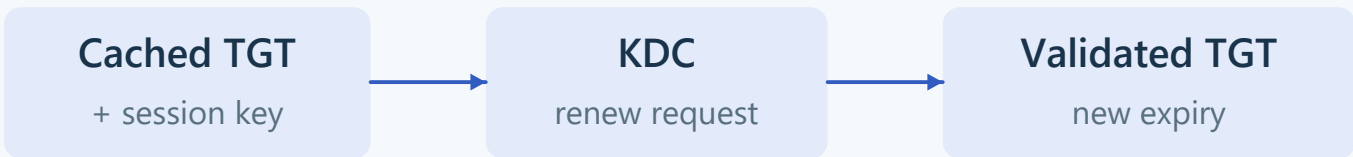
Returns when usable credentials and background maintenance are ready.

01 / RENEW

Use the current ticket

When a renewable TGT has $\leq 20\%$ of its lifetime left (capped at 2 h), and its expiry can extend.

A renewal error retries; no immediate fresh authentication.



Renew-till never moves later.
Retain usable service tickets.

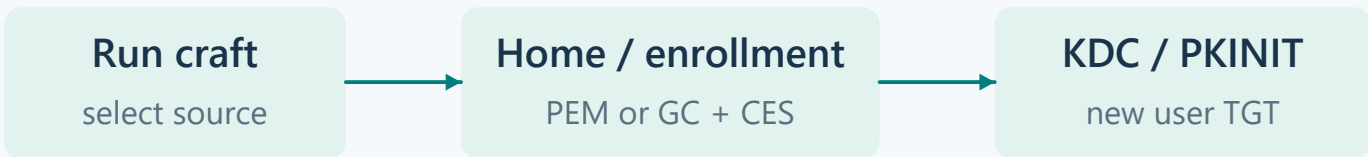
A due renewal takes priority during rollover, keeping usable credentials alive while fresh authentication is delayed.

02 / FRESH AUTHENTICATION

Obtain fresh credentials

Missing or expired TGT; renewal deadline is near; or a ticket that cannot extend is nearing expiry.

Rollover margin: $1/7$ of the renewal window, capped at 1 day.



Reuse a valid home pair, or enroll.
New renewal window; replace cache.

Successful refresh

Publish atomically to .krb5cc_craft (caller-owned, mode 0600).

Refresh failure

Keep the existing cache and retry; the job continues.

RETRY

Renewal: ~30 s to 5 min + jitter.
Fresh acquisition: 1 min to 1 h cooldown.

STOP

Watched PID exits, duration ends or signal.
Maintainer stops; the shared cache remains.

APPLICATION

Reload credentials when authenticating again.
Access follows the user's service permissions.

Home certificate PKINIT without privilege elevation

Ordinary executables run as the caller. The KDC maps the supplied certificate to the fixed Linux-name principal.

